

		Frequency = $c/\lambda = \frac{c\sqrt{2meV}}{h}$
30	C	Paper stops alpha particles, so $\alpha + \beta + background = 352\text{min}^{-1}$ $\beta + background = 256\text{min}^{-1}$ Initial count of beta particle source = $256 - 16 = 240\text{min}^{-1}$ Initial count of alpha particle source = $352 - 256 = 96\text{min}^{-1}$ After 12 days, alpha particle source undergo 3 half-lives and beta particle source undergo 4 half-lives $\text{Final count of alpha particle source} = \frac{96}{2^3} = 12\text{min}^{-1}$ $\text{Final count of beta particle source} = \frac{240}{2^4} = 15\text{min}^{-1}$ Total count = $12 + 15 + 16 = 43\text{min}^{-1}$